

Mechatronics

Mechatronics (MEE4033) · Department of Mechanical Engineering

Lecture 8

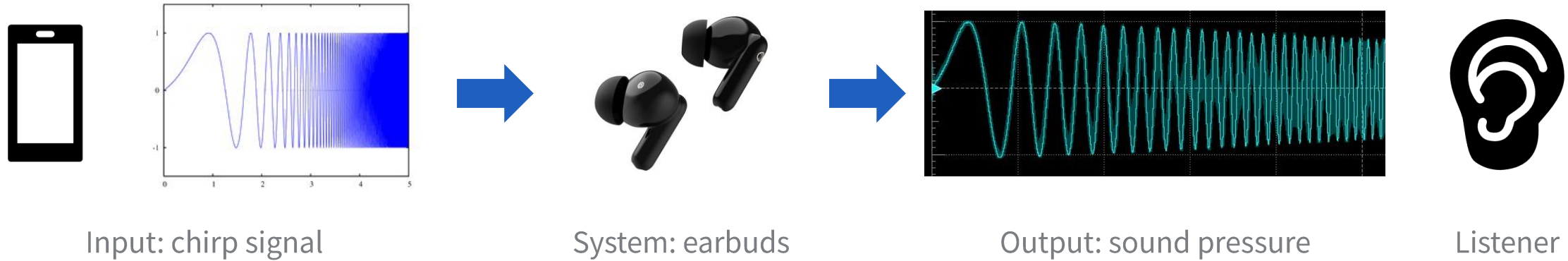
Frequency Response and System Concepts

Principles and Applications of Electrical Engineering, 7th ed., Chapter 5
Electric Circuits, 11th ed., Chapters 12, 13, 14



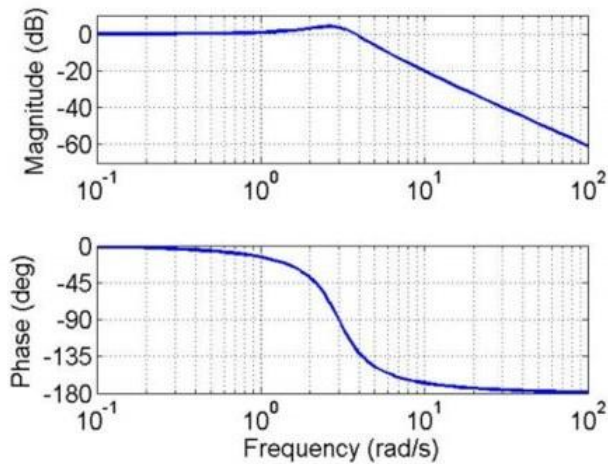
Why Frequency Response?

A phone plays a frequency sweep (chirp) through earbuds. How loud each frequency reaches the ear depends on the frequency.



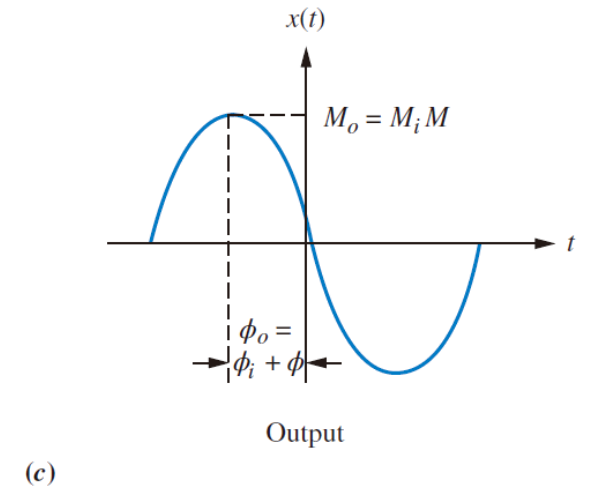
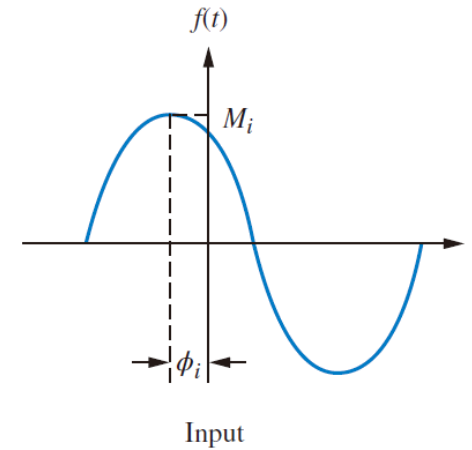
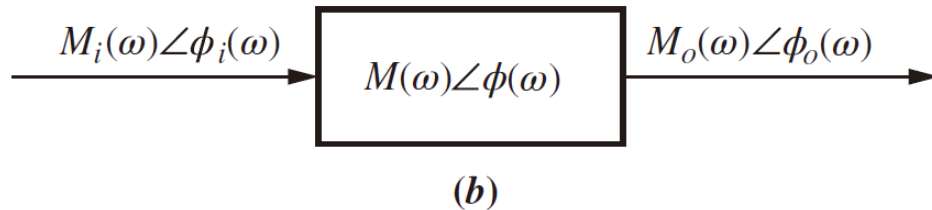
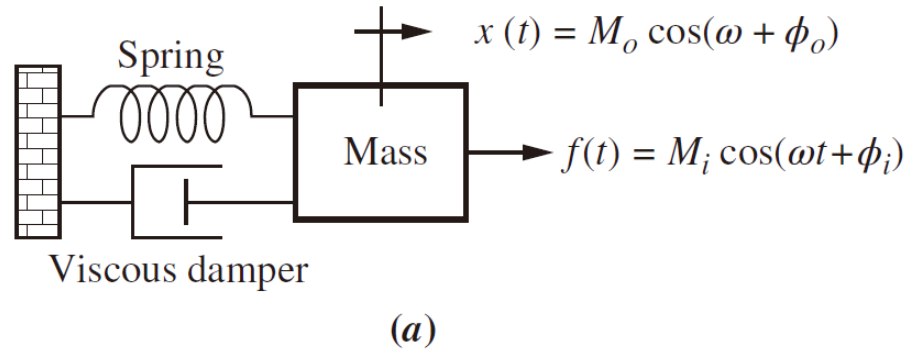
Frequency response

- **Magnitude:** how much each frequency is amplified or attenuated
- **Phase:** how much each frequency is shifted in time
- Plotted against log frequency (Bode plot)



Sinusoidal Response of a Mechanical System

$$M_o(\omega) \angle \phi_o(\omega) = M_i(\omega) M(\omega) \angle [\phi_i(\omega) + \phi(\omega)]$$



In steady state, a linear system driven by a sinusoid responds with a sinusoid of the same frequency. Only the amplitude ($M_o = M_i M$) and the phase ($\phi_o = \phi_i + \phi$) change.

What Frequency Response Measures

- Measures how a circuit responds to sinusoidal inputs of arbitrary frequency
- Lets us compute the output signal for a given input signal
- Defined as the ratio of a chosen output to a chosen input
 - Input: often an independent voltage or current source
 - Output: any voltage or current elsewhere in the circuit
- Frequency response functions are written as G (dimensionless gain) or H (impedance or admittance)

Four Types of Frequency Response

Four distinct types:

$$\mathbf{G}_V(j\omega) = \frac{\mathbf{V}_o(j\omega)}{\mathbf{V}_{in}(j\omega)}$$

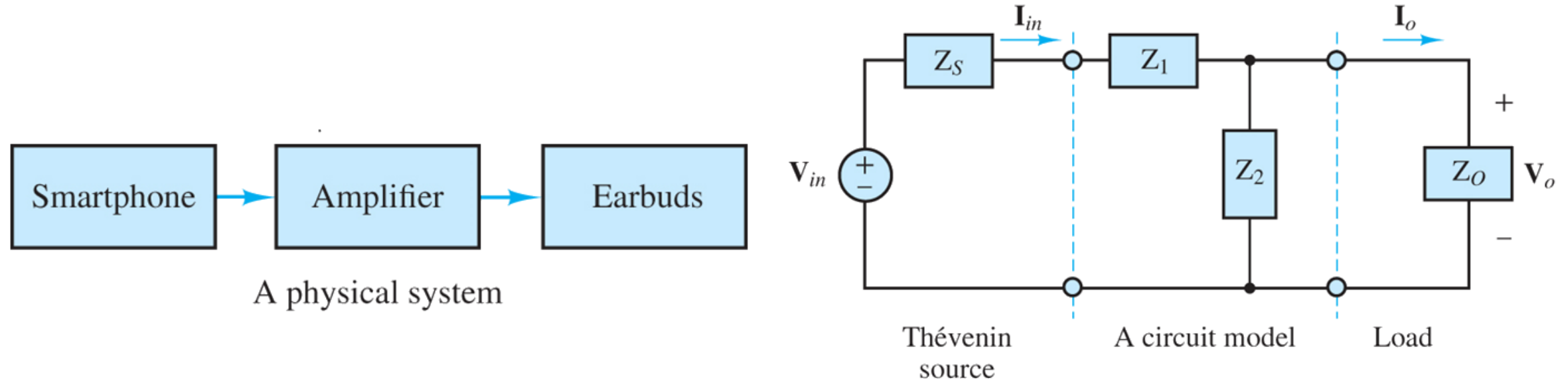
$$\mathbf{G}_I(j\omega) = \frac{\mathbf{I}_o(j\omega)}{\mathbf{I}_{in}(j\omega)}$$

$$\mathbf{H}_Z(j\omega) = \frac{\mathbf{V}_o(j\omega)}{\mathbf{I}_{in}(j\omega)}$$

$$\mathbf{H}_Y(j\omega) = \frac{\mathbf{I}_o(j\omega)}{\mathbf{V}_{in}(j\omega)}$$

- $\mathbf{V}_{in}$ and $\mathbf{I}_{in}$ are often chosen to be independent voltage and current sources
- Outputs $\mathbf{V}_o$ and $\mathbf{I}_o$ are freely chosen and represent the load in a circuit
 - $\mathbf{G}_V, \mathbf{G}_I$: same units in and out (gain). $\mathbf{H}_Z$: V/I (Ω). $\mathbf{H}_Y$: I/V (S)

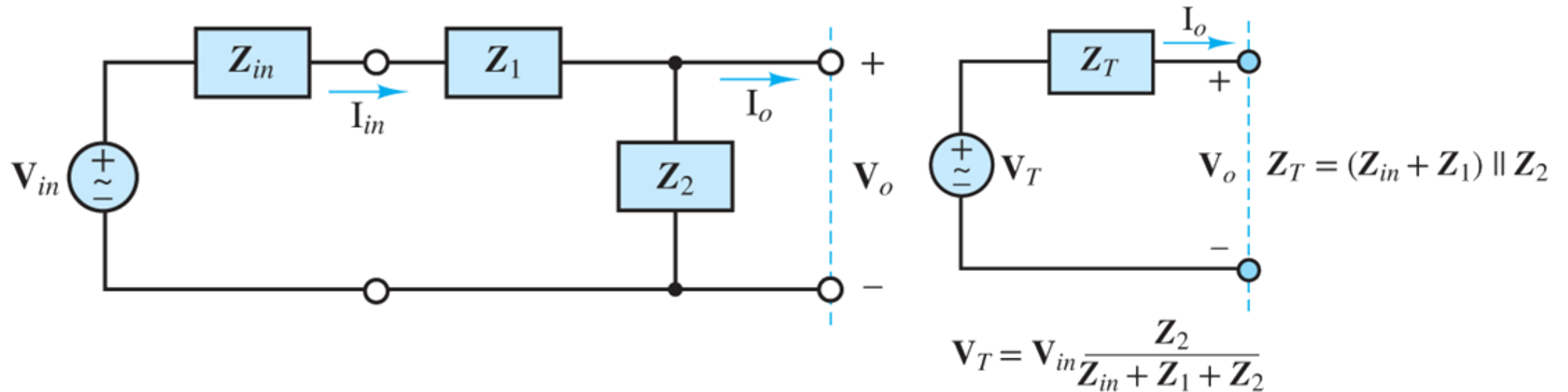
From a Physical System to a Circuit Model



- Smartphone, amplifier, and earbuds form a physical signal chain
- Circuit model: a Thévenin source, a two-port circuit (Z_1, Z_2), and a load Z_o

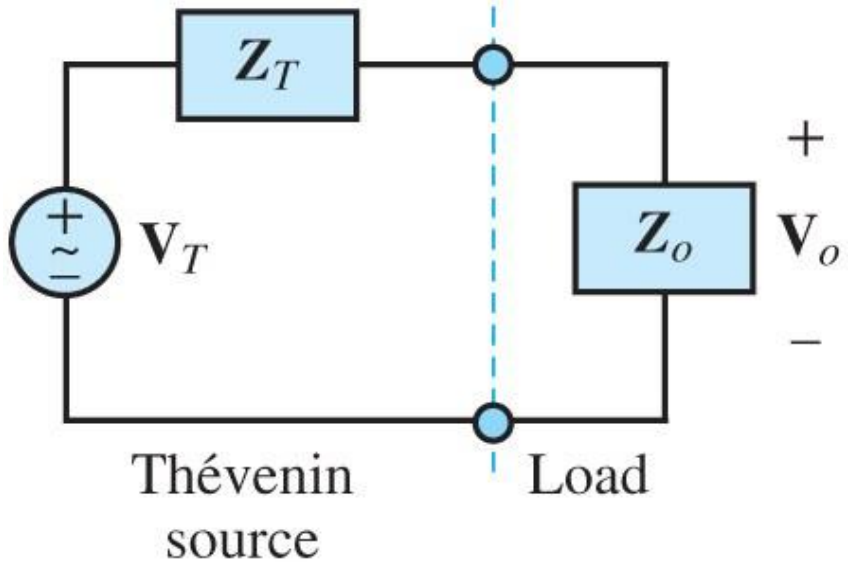
Circuit Simplification (1): Load and Source

- First step: divide the circuit into a load and a source
- The network attached to the load can be replaced with its Thévenin equivalent



Circuit Simplification (2): Voltage Division

Equivalent circuit from the perspective of the load:



$$\begin{aligned}
 V_o &= \frac{Z_o}{Z_o + Z_T} V_T \\
 &= \frac{Z_o}{Z_o + (Z_{in} + Z_1)Z_2 / (Z_{in} + Z_1 + Z_2)} \cdot \frac{Z_2}{Z_{in} + Z_1 + Z_2} V_{in} \\
 &= \frac{Z_o Z_2}{Z_o(Z_{in} + Z_1 + Z_2) + (Z_{in} + Z_1)Z_2} V_{in}
 \end{aligned}$$

- Use voltage division to express V_o in terms of V_T
- Then substitute V_T and Z_T to express V_o in terms of V_{in}

Gain from the Circuit Impedances

- The gain is known once the circuit impedances are known

$$\mathbf{G}_v(j\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_{in}}(j\omega) = \frac{\mathbf{Z}_o \mathbf{Z}_2}{\mathbf{Z}_o (\mathbf{Z}_{in} + \mathbf{Z}_1 + \mathbf{Z}_2) + (\mathbf{Z}_{in} + \mathbf{Z}_1) \mathbf{Z}_2}$$

- $V_o(j\omega)$ is a phase-shifted and amplitude-scaled version of $V_{in}(j\omega)$

Magnitude and Phase of the Load Voltage

Phasor load voltage in terms of the phasor source voltage:

$$\begin{aligned}\mathbf{V}_o(j\omega) &= \mathbf{G}_V(j\omega) \cdot \mathbf{V}_{in}(j\omega) \\ V_o e^{j\phi_o} &= |\mathbf{G}_V| e^{j\angle \mathbf{G}_V} \cdot V_{in} e^{j\phi_{in}}\end{aligned}$$

Such that:

$$V_o = |\mathbf{G}_V| \cdot V_{in} \qquad \phi_o = \angle \mathbf{G}_V + \phi_{in}$$

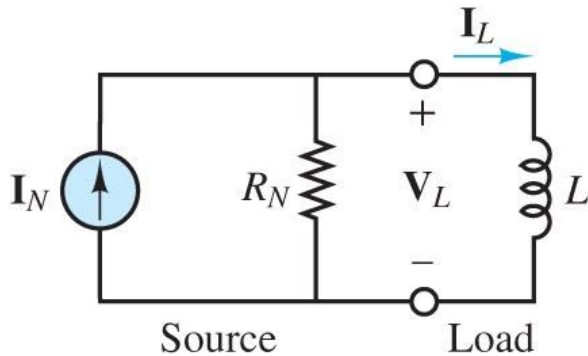
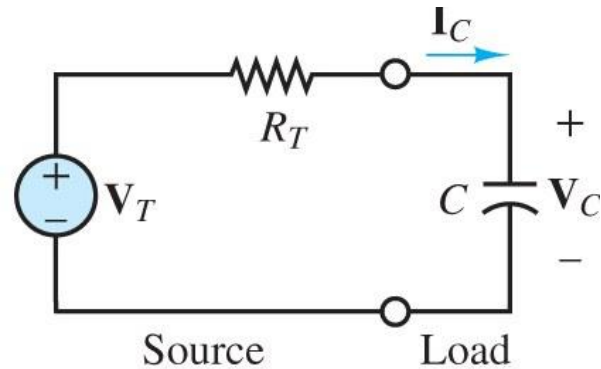
At a given angular frequency ω :

- The load voltage is a sinusoid with the same frequency as the source voltage
 - Amplitude scaled by $|\mathbf{G}_V|$, phase shifted by $\angle \mathbf{G}_V$

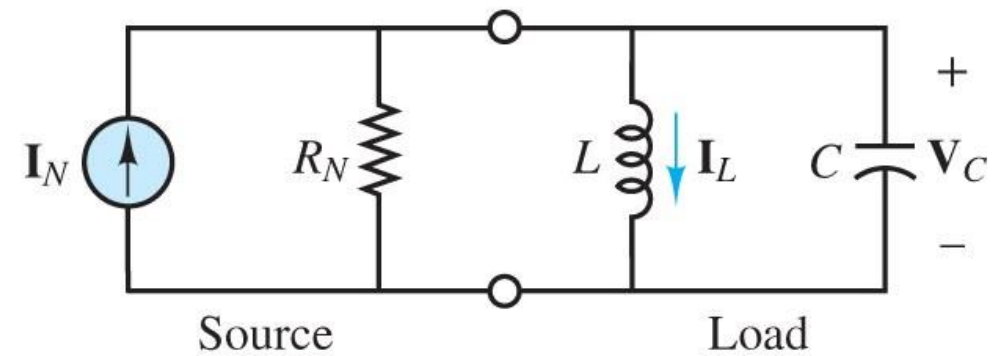
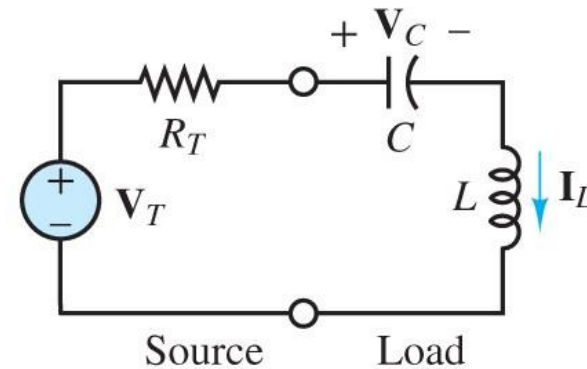
First- and Second-Order Archetypes

- If a circuit is first-order, or second-order with the storage elements in series or in parallel, it can be simplified to one of four archetypes

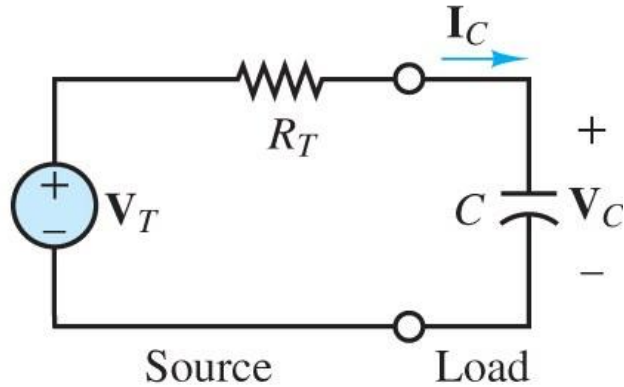
First-order



Second-order



First-Order Circuit with One Capacitor



- Ohm's law:

$$\mathbf{I}_C = \frac{\mathbf{V}_T}{R_T + \mathbf{Z}_C}$$

- Impedance of C:

$$\mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

- FRF between $\mathbf{I}_C$ and $\mathbf{V}_T$:

$$\mathbf{H}_Y(j\omega) = \frac{\mathbf{I}_C}{\mathbf{V}_T} = \frac{(j\omega)C}{1 + (j\omega)\tau} \quad (\tau = R_T C)$$

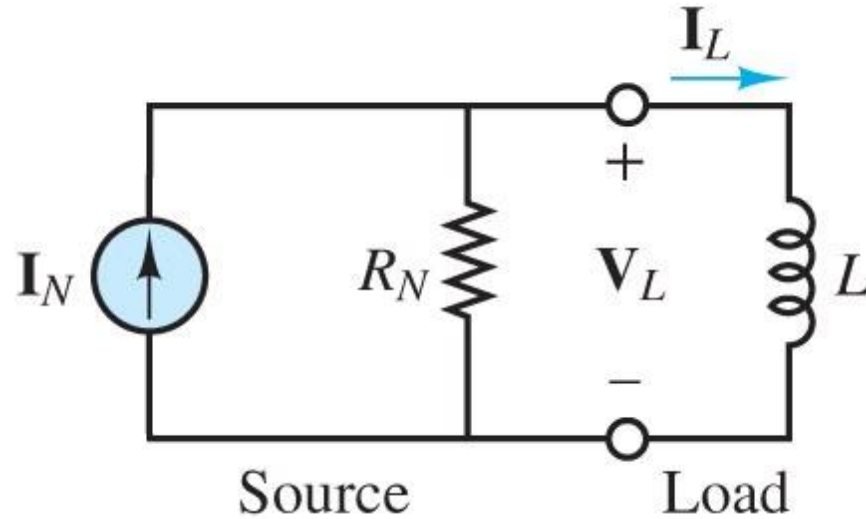
- FRF between $\mathbf{V}_C$ and $\mathbf{V}_T$:

$$\mathbf{G}_V(j\omega) = \frac{\mathbf{V}_C}{\mathbf{V}_T} = \frac{\mathbf{I}_C}{\mathbf{V}_T} \mathbf{Z}_C = \frac{(j\omega)C}{1 + (j\omega)\tau} \cdot \frac{1}{(j\omega)C} = \frac{1}{1 + (j\omega)\tau}$$

Different numerators: they reflect which circuit variable is chosen as the output

Common denominator: it expresses the characteristic dynamics of the circuit

First-Order Circuit with One Inductor

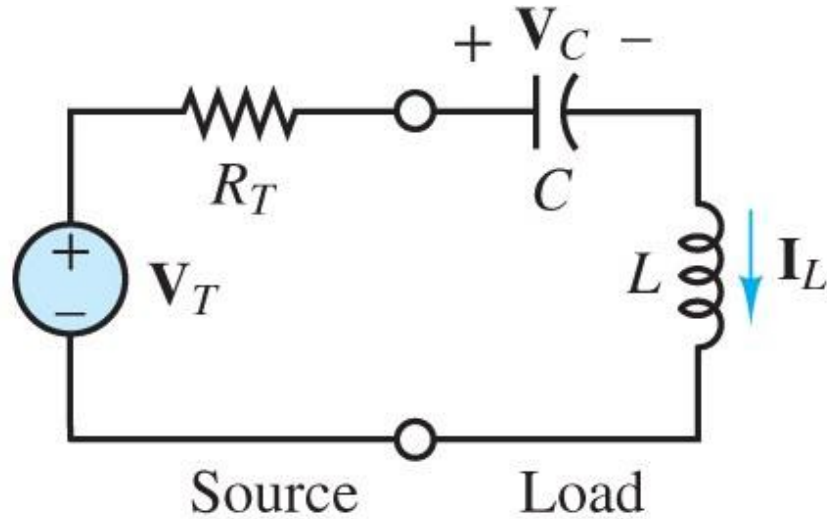


$$\mathbf{G}_I(j\omega) = \frac{\mathbf{I}_L}{\mathbf{I}_N} = \frac{\mathbf{V}_L}{\mathbf{I}_N} \frac{1}{\mathbf{Z}_L} = \frac{1}{1 + (j\omega)\tau}$$

$$\tau = L / R_N$$

Current divider between R_N and L

Second-Order Series LC Circuit

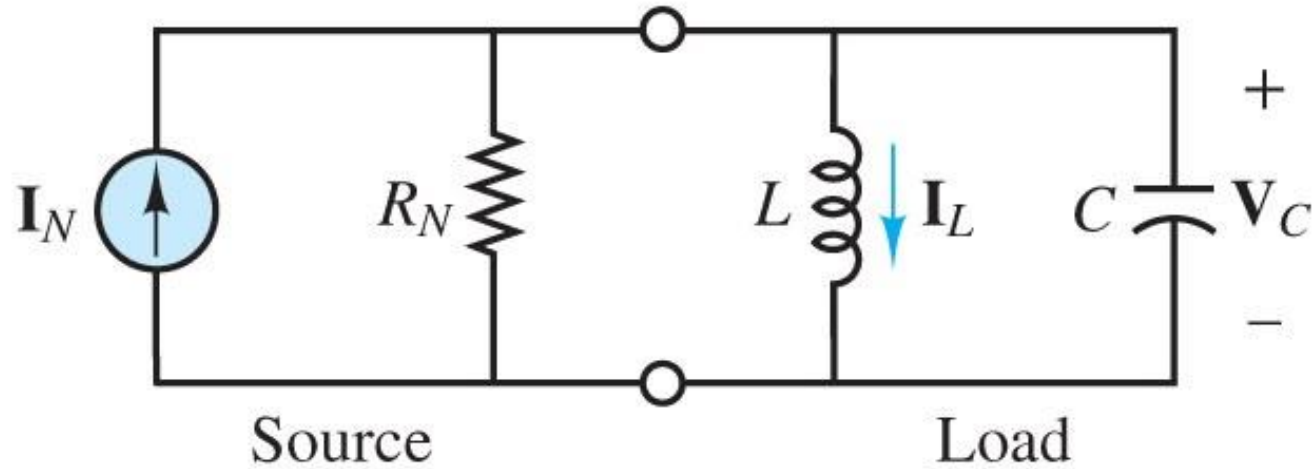


$$\mathbf{G}_V(j\omega) = \frac{\mathbf{I}_L}{\mathbf{V}_T} \mathbf{Z}_C = \mathbf{H}_Y \mathbf{Z}_C = \frac{1}{1 + (j\omega)\tau + (j\omega/\omega_n)^2}$$

$$\tau = R_T C \quad \omega_n^2 = 1/LC$$

τ : time constant, ω_n : natural frequency

Second-Order Parallel LC Circuit



$$\mathbf{H}_Z(j\omega) = \frac{(j\omega)L}{1 + (j\omega)L/R_N + (j\omega)^2 LC} = \frac{(j\omega)L}{1 + (j\omega)\tau + (j\omega/\omega_n)^2}$$

$$\tau = L/R_N \quad \omega_n^2 = 1/LC$$

Transfer impedance: V_C / I_N

Why the Laplace Transform?

- We used the phasor transform to find the steady-state response of a circuit to a sinusoidal source
- In many cases we need the complete response of a circuit to a sinusoidal source
- The complete response has two parts:
 - 1) the steady-state response, which has the same form as the input
 - 2) the transient response, which decays to zero as time progresses
- The Laplace transform finds the complete response of a circuit to a sinusoidal source (and to other inputs)

Definition of the Laplace Transform

Definition of the Laplace transform:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

- Laplace transforms do not exist for some functions, but those functions are “pathological” and not generally of interest to engineers

Laplace Transforms of Common Functions

Use the definition of the Laplace transform to calculate the transforms of some functions of interest:

$$\mathcal{L}\{\delta(t)\} = \int_0^{\infty} \delta(t) e^{-st} dt = \int_0^{\infty} \delta(t-0) e^{-st} dt = e^{-s(0)} = 1$$

$$\mathcal{L}\{u(t)\} = \int_0^{\infty} u(t) e^{-st} dt = \int_0^{\infty} 1 e^{-st} dt = \frac{1}{-s} e^{-st} \Big|_0^{\infty} = 0 - \frac{1}{-s} = \frac{1}{s}$$

$$\mathcal{L}\{e^{-at}\} = \int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt = \frac{1}{-(s+a)} e^{-(s+a)t} \Big|_0^{\infty} = 0 - \frac{1}{-(s+a)} = \frac{1}{(s+a)}$$

$$\begin{aligned} \mathcal{L}\{\sin \omega t\} &= \int_0^{\infty} \left[\frac{e^{j\omega t} - e^{-j\omega t}}{2j} \right] e^{-st} dt = \frac{1}{j2} \int_0^{\infty} [e^{-(s-j\omega)t} - e^{-(s+j\omega)t}] dt \\ &= \frac{1}{j2} \left[\frac{e^{-(s-j\omega)t}}{-(s-j\omega)} \right]_0^{\infty} - \frac{1}{j2} \left[\frac{e^{-(s+j\omega)t}}{-(s+j\omega)} \right]_0^{\infty} = \frac{1}{j2} \left[\frac{1}{(s-j\omega)} - \frac{1}{(s+j\omega)} \right] = \frac{\omega}{s^2 + \omega^2} \end{aligned}$$

Laplace Transform Pairs (Table 12.1)

TABLE 12.1 An Abbreviated List of Laplace Transform Pairs

Type	$f(t)$ ($t > 0^-$)	$F(s)$
(impulse)	$\delta(t)$	1
(step)	$u(t)$	$\frac{1}{s}$
(ramp)	t	$\frac{1}{s^2}$
(exponential)	e^{-at}	$\frac{1}{s + a}$
(sine)	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
(cosine)	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
(damped ramp)	te^{-at}	$\frac{1}{(s + a)^2}$
(damped sine)	$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
(damped cosine)	$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

Operational Transforms

Using the definition, determine the effect of various operations on time-domain functions when the result is Laplace-transformed. These are collected in the Operational Transform table (Table 12.2).

Linearity

$$\begin{aligned}\mathcal{L}\{K_1 f_1(t) + K_2 f_2(t) - K_3 f_3(t)\} &= \int_0^{\infty} [K_1 f_1(t) + K_2 f_2(t) - K_3 f_3(t)] e^{-st} dt \\ &= K_1 F_1(s) + K_2 F_2(s) - K_3 F_3(s)\end{aligned}$$

Differentiation

(integration by parts)

$$\begin{aligned}\mathcal{L}\left\{\frac{df(t)}{dt}\right\} &= \left[e^{-st} f(t)\right]_0^{\infty} - \int_0^{\infty} f(t) (-se^{-st}) dt \\ &= -f(0) + s \int_0^{\infty} f(t) e^{-st} dt = sF(s) - f(0)\end{aligned}$$

Integration

(integration by parts)

$$\mathcal{L}\left\{\int_0^t f(x) dx\right\} = \underbrace{\left[-\frac{e^{-st}}{s} \int_0^t f(x) dx\right]_0^{\infty}}_{=0} + \frac{1}{s} \int_0^{\infty} f(t) e^{-st} dt = \frac{F(s)}{s}$$

Used on the next slide: the KCL term $(1/L) \int_0^t v(x) dx$ becomes $V(s)/(Ls)$.

Example: Parallel RLC Circuit in the s-Domain

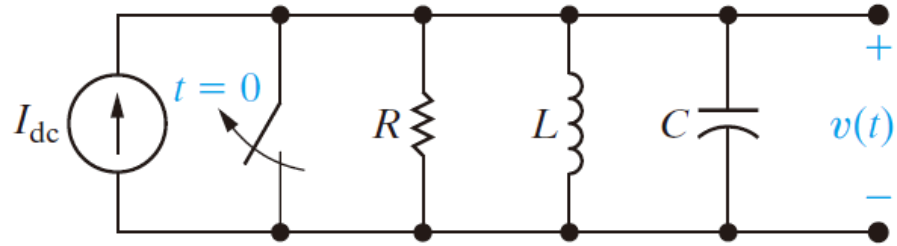


Figure 12.16 ▲ A parallel RLC circuit.

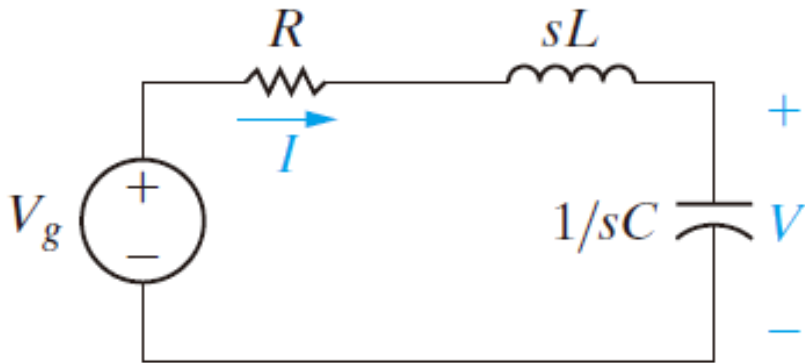
- KCL: $\frac{v(t)}{R} + \frac{1}{L} \int_0^t v(x) dx + C \frac{dv(t)}{dt} = I_{dc}u(t)$
- Laplace transform: $\frac{V(s)}{R} + \frac{1}{L} \frac{V(s)}{s} + C[sV(s) - v(0^-)] = I_{dc} \left(\frac{1}{s} \right)$
- Solve for the output:
$$V(s) = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}$$
 - Output: $V(s)$ Input: I_{dc} (step at $t = 0$) No initial condition: $v(0^-) = 0$

Transfer Function

- **Definition:** the s-domain ratio of the Laplace transform of the output (response) to the Laplace transform of the input (source)

DEFINITION OF A TRANSFER FUNCTION

$$H(s) = \frac{Y(s)}{X(s)},$$



$$H(s) = \frac{I}{V_g} = \frac{1}{R + sL + 1/sC} = \frac{sC}{s^2 LC + RCs + 1}$$

$$H(s) = \frac{V}{V_g} = \frac{1/sC}{R + sL + 1/sC} = \frac{1}{s^2 LC + RCs + 1}$$

Inverse Laplace Transform: Poles and Zeros

Inverse Laplace transform using partial fraction expansion of $F(s) = N(s)/D(s)$:

- The roots of $D(s)$ (values of s that make $D(s) = 0$) are called poles
- The roots of $N(s)$ (values of s that make $N(s) = 0$) are called zeros

Example:

$$F(s) = \frac{40s + 360}{s^3 + 14s^2 + 24s} = \frac{40(s + 9)}{s(s + 2)(s + 12)}$$

Zero: $s = -9$ Poles: $s = 0, -2, -12$

Partial Fraction Example

$$F(s) = \frac{40s + 360}{s(s+2)(s+12)} = \frac{K_1}{s} + \frac{K_2}{s+2} + \frac{K_3}{s+12}$$

Each coefficient: multiply by its denominator factor and evaluate at that pole

$$K_1 = \left. \frac{40s + 360}{(s+2)(s+12)} \right|_{s=0} = 15; \quad K_2 = \left. \frac{40s + 360}{s(s+12)} \right|_{s=-2} = -14; \quad K_3 = \left. \frac{40s + 360}{s(s+2)} \right|_{s=-12} = -1$$

$$\therefore I_1(s) = \frac{15}{s} + \frac{-14}{s+2} + \frac{-1}{s+12}$$

$$i_1(t) = \mathcal{L}^{-1} \left\{ \frac{15}{s} + \frac{-14}{s+2} + \frac{-1}{s+12} \right\} = [15 - 14e^{-2t} - e^{-12t}]u(t) \text{ A}$$

Partial Fractions: Distinct Real Roots

Partial fractions for distinct real roots of $D(s)$ (denominator):

$$F(s) = \frac{N(s)}{(s + p_1)(s + p_2)(s + p_3)} = \frac{K_1}{s + p_1} + \frac{K_2}{s + p_2} + \frac{K_3}{s + p_3}$$

– Multiply both sides by $(s + p_1)$:

$$\frac{N(s)}{(s + p_2)(s + p_3)} = K_1 + \frac{K_2(s + p_1)}{s + p_2} + \frac{K_3(s + p_1)}{s + p_3}$$

– Evaluate at $s = -p_1$: the K_2 and K_3 terms vanish because of the factor $(s + p_1)$

$$K_1 = \left. \frac{N(s)}{(s + p_2)(s + p_3)} \right|_{s=-p_1} = (s + p_1)F(s)|_{s=-p_1}$$

Initial and Final Value Theorems

- Check the initial and final values in the s-domain before inverse-transforming, using the Initial Value Theorem (IVT) and the Final Value Theorem (FVT)

Initial value theorem:

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s)$$

- Valid only if $f(t)$ contains no impulse functions

Final value theorem:

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

- Valid only if all poles of $F(s)$, except possibly a single pole at the origin, lie in the left half of the s-plane

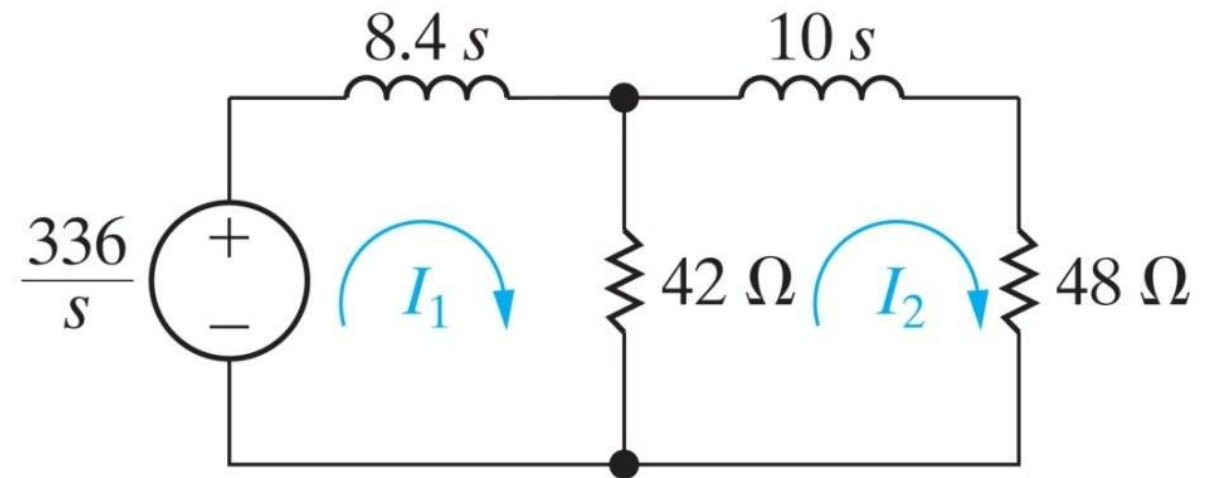
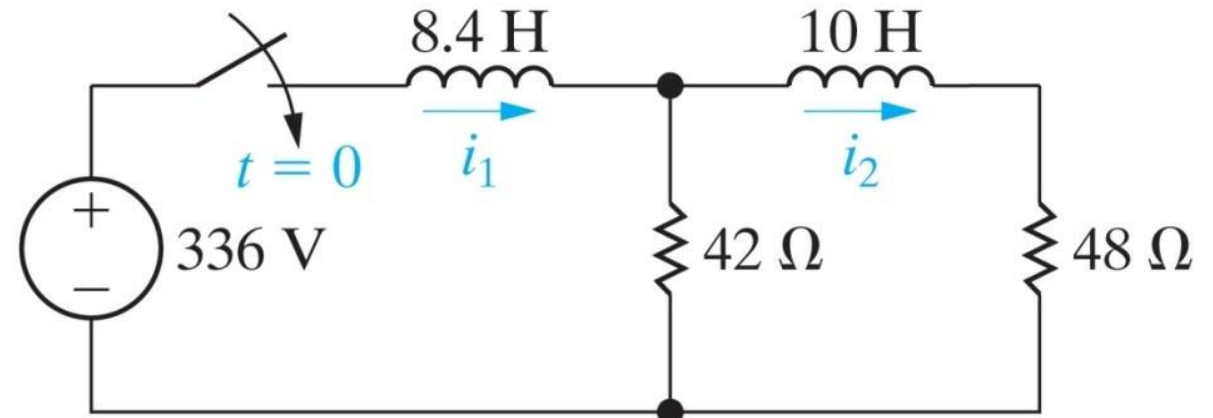
Example: Two-Mesh RL Circuit

There is no initial energy stored in this circuit. Find $i_1(t)$ and $i_2(t)$ for $t > 0$.

$$I_1(s) = \frac{40s + 360}{s^3 + 14s^2 + 24s}$$

$$I_2(s) = \frac{168}{s^3 + 14s^2 + 24s}$$

Check your answers using the IVT and the FVT.

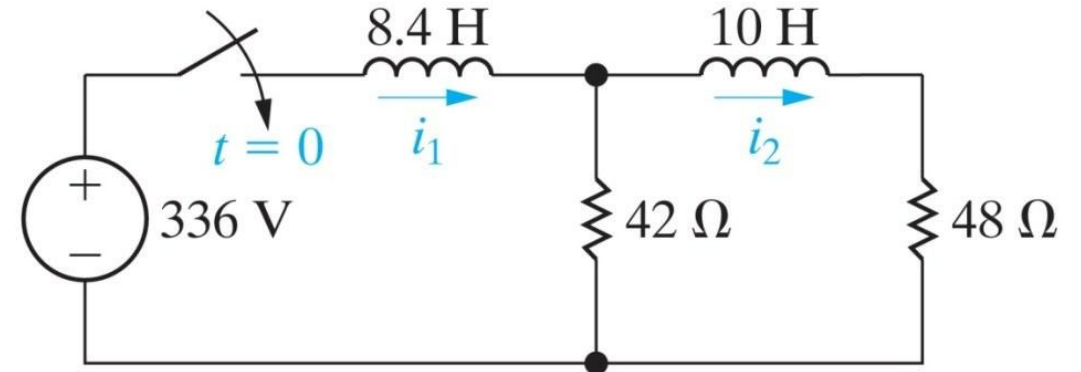


s-domain circuit with mesh currents I_1 and I_2

Checking with the Initial Value Theorem

From the circuit: $i_1(0^+) = i_2(0^+) = 0$

- Inductor currents cannot change instantly



$$I_1(s) = \frac{40s + 360}{s^3 + 14s^2 + 24s}$$

$$\begin{aligned} \lim_{t \rightarrow 0^+} i_1(t) &= \lim_{s \rightarrow \infty} sI_1(s) = \lim_{s \rightarrow \infty} \frac{40s^2 + 360s}{s^3 + 14s^2 + 24s} \\ &= \lim_{s \rightarrow \infty} \frac{40/s + 360/s^2}{1 + 14/s + 24/s^2} = 0 \text{ A } \checkmark \end{aligned}$$

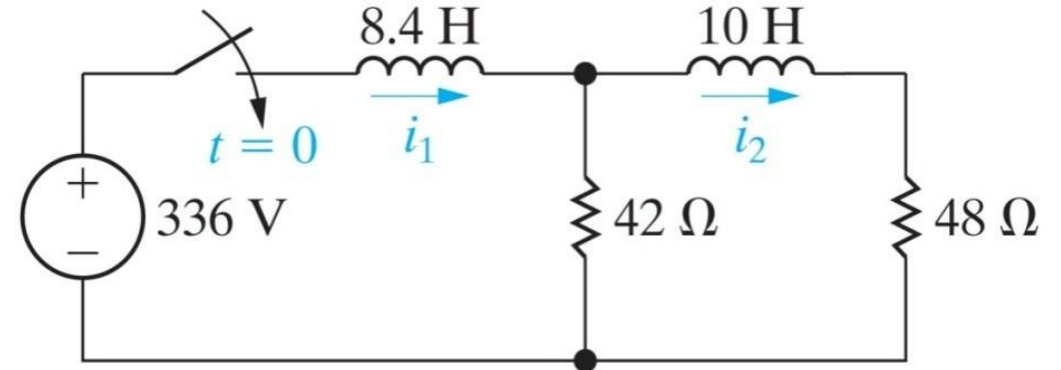
$$I_2(s) = \frac{168}{s^3 + 14s^2 + 24s}$$

$$\begin{aligned} \lim_{t \rightarrow 0^+} i_2(t) &= \lim_{s \rightarrow \infty} sI_2(s) = \lim_{s \rightarrow \infty} \frac{168s}{s^3 + 14s^2 + 24s} \\ &= \lim_{s \rightarrow \infty} \frac{168/s^2}{1 + 14/s + 24/s^2} = 0 \text{ A } \checkmark \end{aligned}$$

Checking with the Final Value Theorem

From the circuit: $i_1(\infty) = 15 \text{ A}$, $i_2(\infty) = 7 \text{ A}$

- Inductors act as short circuits in DC steady state



$$\begin{aligned} \lim_{t \rightarrow \infty} i_1(t) &= \lim_{s \rightarrow 0} sI_1(s) = \lim_{s \rightarrow 0} \frac{40s^2 + 360s}{s^3 + 14s^2 + 24s} \\ &= \lim_{s \rightarrow 0} \frac{40s + 360}{s^2 + 14s + 24} = \frac{360}{24} = 15 \text{ A} \checkmark \end{aligned}$$

$$\begin{aligned} \lim_{t \rightarrow \infty} i_2(t) &= \lim_{s \rightarrow 0} sI_2(s) = \lim_{s \rightarrow 0} \frac{168s}{s^3 + 14s^2 + 24s} \\ &= \lim_{s \rightarrow 0} \frac{168}{s^2 + 14s + 24} = \frac{168}{24} = 7 \text{ A} \checkmark \end{aligned}$$

- FVT applies: the poles of $sI_1(s)$ and $sI_2(s)$ are $s = -2$ and -12 , both in the left half-plane

Sinusoidal Input and the Transfer Function

- Measures how a circuit responds (output) to sinusoidal inputs of arbitrary frequency

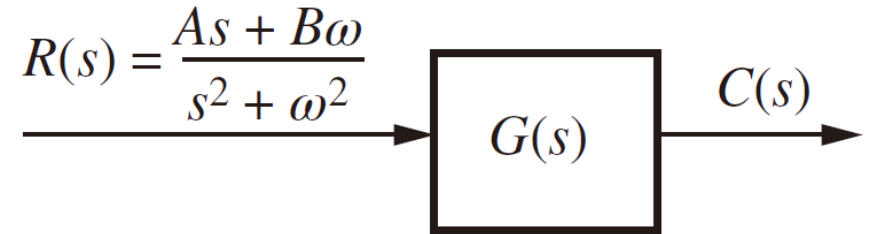
Time domain

$$r(t) = A \cos \omega t + B \sin \omega t = \sqrt{A^2 + B^2} \cos[\omega t - \tan^{-1}(B/A)]$$

Phasor form

$$M_i \angle \phi_i, \quad M_i = \sqrt{A^2 + B^2}, \quad \phi_i = -\tan^{-1}(B/A)$$

$$\begin{array}{ccccc} A - jB & = & M_i \angle \phi_i & = & M_i e^{j\phi_i} \\ \text{rectangular} & & \text{polar} & & \text{Euler} \end{array}$$



$$C(s) = \frac{As + B\omega}{s^2 + \omega^2} G(s)$$

Sinusoidal input in, $C(s)$ out: how is the output calculated?

Partial Fraction Expansion of the Output

$$C(s) = \frac{As + B\omega}{(s + j\omega)(s - j\omega)} G(s)$$

$$= \frac{K_1}{s + j\omega} + \frac{K_2}{s - j\omega} + \text{Partial fraction terms from } G(s)$$

$$K_1 = \left. \frac{As + B\omega}{s - j\omega} G(s) \right|_{s \rightarrow -j\omega} = \frac{1}{2} (A + jB) G(-j\omega) = \frac{1}{2} M_i e^{-j\phi_i} M_G e^{-j\phi_G}$$

$$= \frac{M_i M_G}{2} e^{-j(\phi_i + \phi_G)}$$

$$K_2 = \left. \frac{As + B\omega}{s + j\omega} G(s) \right|_{s \rightarrow +j\omega} = \frac{1}{2} (A - jB) G(j\omega) = \frac{1}{2} M_i e^{j\phi_i} M_G e^{j\phi_G}$$

$$= \frac{M_i M_G}{2} e^{j(\phi_i + \phi_G)} = K_1^*$$

Terms

- K_1^* is the complex conjugate of K_1
- $M_G = |G(j\omega)|$
- $\phi_G = \angle G(j\omega)$
- The remaining terms come from the poles of $G(s)$

Steady-State Response

- The steady-state response is the part of the partial-fraction expansion that comes from the input poles ($s = \pm j\omega$), that is, the first two terms

$$C_{ss}(s) = \frac{K_1}{s + j\omega} + \frac{K_2}{s - j\omega}$$

$$C_{ss}(s) = \frac{\frac{M_i M_G}{2} e^{-j(\phi_i + \phi_G)}}{s + j\omega} + \frac{\frac{M_i M_G}{2} e^{j(\phi_i + \phi_G)}}{s - j\omega}$$

Taking the inverse Laplace transform:

$$\begin{aligned} c(t) &= M_i M_G \left(\frac{e^{-j(\omega t + \phi_i + \phi_G)} + e^{j(\omega t + \phi_i + \phi_G)}}{2} \right) \\ &= M_i M_G \cos(\omega t + \phi_i + \phi_G) \end{aligned}$$

Phasor form

$$\underbrace{M_o \angle \phi_o}_{\text{output}} = \underbrace{(M_i \angle \phi_i)}_{\text{input}} \underbrace{(M_G \angle \phi_G)}_{\text{frequency response function}}$$

If $G(s)$ is stable, the terms from its poles decay to zero (transient response)

Frequency Response Function $G(j\omega)$

Recall that:

$$\underbrace{M_o \angle \phi_o}_{\text{output}} = \underbrace{(M_i \angle \phi_i)}_{\text{input}} \underbrace{(M_G \angle \phi_G)}_{\text{frequency response function}}$$

$$M_G = |G(j\omega)|, \quad \phi_G = \angle G(j\omega)$$

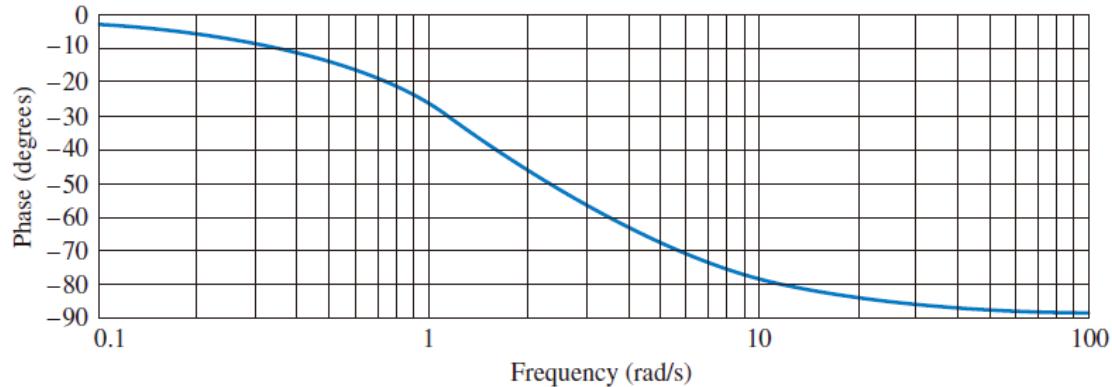
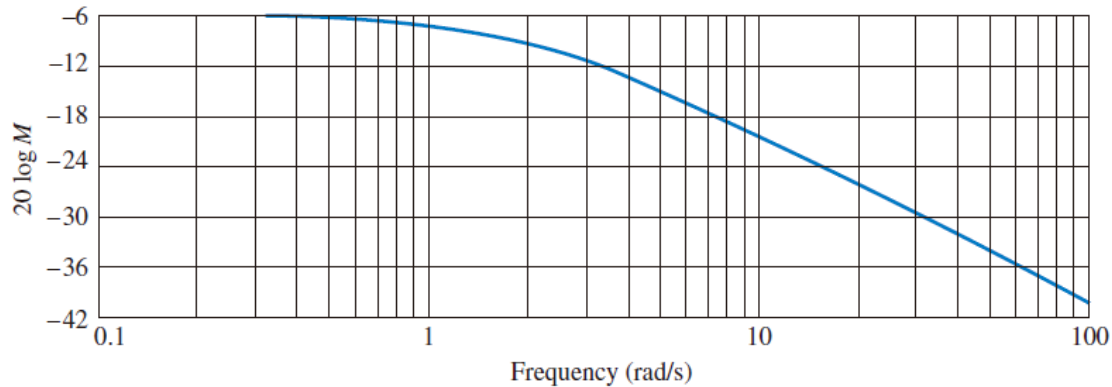
- The frequency response of a system whose transfer function is $G(s)$ is:

$$G(j\omega) = G(s)|_{s=j\omega}$$

Evaluate the transfer function on the imaginary axis, $s = j\omega$

Example: Bode Plot of $G(s) = 1/(s + 2)$

$$G(s) = \frac{1}{s + 2} \Rightarrow G(j\omega) = \frac{1}{2 + j\omega}$$



- Low frequency ($\omega \ll 2$): $|G| \rightarrow 1/2$ (-6 dB), phase $\rightarrow 0^\circ$
- Corner frequency $\omega = 2$ rad/s: $|G| = 1/\sqrt{8}$ (≈ -9 dB), phase = -45°
- High frequency ($\omega \gg 2$): magnitude falls at -20 dB/decade, phase $\rightarrow -90^\circ$